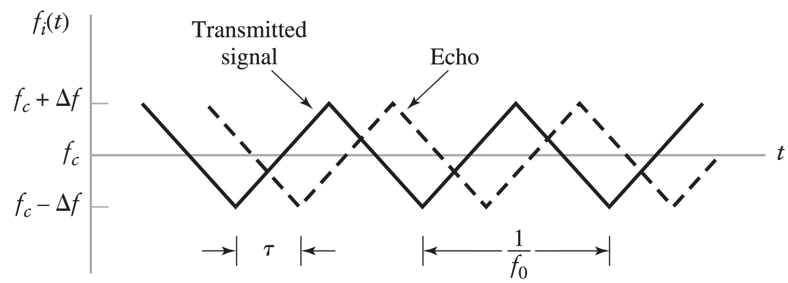
EE121 Fall 2025 Miki Lustig
Digital Communications Systems

Assignment 5

Due Sunday October 12th, 2025, Self Grading Due Sunday October 19th, 2025

1. Madhow Problem 3.4
2. Madhow Problem 3.16
3. In a *frequency-modulated radar* the instantaneous frequency of the transmitted carrier is varied as shown below. Such a signal is generated by frequency modulation with a periodic triangular modulating wave. The instantaneous frequency of the received echo signal is shown dashed below where τ is the round-trip delay time. The transmitted and received echo signals are applied to a mixer, and the difference frequency component is retained. Assuming that $f_0\tau \ll 1$ for all t, determine the number of beat cycles at the mixer output, averaged over one second, in terms of the peak deviation Δf of the carrier frequency, the delay τ, and the repetition frequency f_0 of the transmitted signal. (The beat refers to a signal whose frequency is the difference between the frequencies of the two input signals.)



4. **Self-grade Homework 4.** Fill out the self-grade rubric for Homework 4. Attach to Homework 5 submission.